Solutions

SAFETY BAY
SENIOR HIGH SCHOOL
imagine believe achieve

**Sem1 Examination 2015**

**Question/answer booklet**

**Name:** ______________________________

**Teacher:** ______________________________

# YEAR 10
# ACM MATHEMATICS

**Section One: 34 Marks**
**Calculator-free**

| |
|---|
| ____ |
| 63 |
| =_____ % |

## Time allowed for this section

**Section One**

Reading time before commencing work: 2 minutes
Working time for paper: 30 minutes

## Material required/recommended for this paper

**To be provided by the supervisor**
This question/answer Booklet
Formula sheet

**To be provided by the student**
*Standard items:* pens, pencils, pencil sharpener, eraser, correction fluid/tape, ruler, highlighter

## Important note to students

This test comprises of **TWO sections**. The **first section** is **calculator free** where no calculators of any kind are to be used. The **second section** is **calculator assumed** where a scientific calculator may be used. All questions must be answered in both sections. **Answers should be rounded appropriately**. All working should be shown in the space provided. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks.

No pen, pencils, highlights etc. may be used during reading time. This time is to be used to read through the assessment and check that you understand what is being asked of you. You may speak with the teacher/supervisor during this time (by putting up your hand and waiting patiently for them to approach you) but you may only ask clarification questions and not how to solve the problems. After reading time has ended, you may not ask any more questions.

1. **[4 Marks: 1, 1, 1, 1]**

**a)** Find two numbers which add to make 22 and multiply to make 40

20 and 2 ✓

**b)** Calculate the value of $12^1$

12 ✓

**c)** Which of the following numbers is probably irrational? Circle your answer.

0.862 323 2… 0.584 597 2

0.111 111 1… 0.010 214 2… ✓

**d)** Which of the following is not a surd? Circle your answer

$\sqrt{49}$ ✓ $\sqrt{5}$

$\sqrt{2}$ $\sqrt{80}$

2. **[6 Marks: 1, 1, 2, 2]**
Factorise the following

**a)** $9x + 18$

$9(x+2)$ ✓

**b)** $x^2 - 5x$

$x(x-5)$ ✓

**c)** $x^2 - 23x - 24$

$(x+1)(x-24)$

✓✓

**d)** $6x^2 - 6x - 36$

$6(x^2-x-6)$ ✓

$6(x+2)(x-3)$ ✓

**3. [8 Marks: 1, 1, 2, 2, 2]**
Expand and simplify the following

**a)** $2(x+5)$

$2x + 10$ ✓

**b)** $5x(8-2x)$

$40x - 10x^2$ ✓

**c)** $3a(a-2b+5c)$

$3a^2 - 6ab + 15ac$

✓✓

(-1 for a Small error)

**d)** $(x+3)(x-3)$

$x^2 - 9$ ✓✓

**e)** $(x-1)^2$

$x^2 - 2x + 1$ ✓✓

**4. [4 Marks]**
Find the surface area of a cube with side length of 5 cm. Show full working out.
hint: draw a diagram.

Area of side $= 5 \times 5$
$= 25\ cm^2$ ✓✓

1 working Mark can be given for a valid diagram.

Surface Area $= 6 \times 25$
$= 150\ cm^2$ ✓✓

**5. [12 Marks: 1, 2, 2, 2, 2, 1, 2]**
Simplify the following:

**a)** $(x^3y^4)^5$

$x^{15}y^{20}$ ✓

**b)** $(x^8)^4 \div (x^6)^2$

or similar

$\frac{x^{32}}{x^{12}}$ ✓ $= x^{20}$ ✓

**c)** $\frac{3x-7}{5} \div \frac{x-2}{8}$

$\frac{8(3x-7)}{5(x-2)}$ ✓✓

**d)** $\frac{d}{2} + \frac{d}{5}$

$\frac{5d+2d}{10}$ ✓ $= \frac{7d}{10}$ ✓

**e)** $\frac{7e}{9f} \times \frac{5f^2}{e}$

$\frac{35ef^2}{9ef} = \frac{35f}{9}$ ✓✓

**f)** $\sqrt{40}$

$\sqrt{4 \times 10}$

$= 2\sqrt{10}$ ✓

**g)** $3\sqrt{17} + 9\sqrt{97} + 7\sqrt{17} - 4\sqrt{97}$

$3\sqrt{17} + 7\sqrt{17} + 9\sqrt{97} - 4\sqrt{97}$

$= 10\sqrt{17} + 5\sqrt{97}$

✓ ✓

**~ END OF SECTION ONE ~**

**6. [4 Marks: 1, 1, 1, 1]**

1 Calculate the values of the following

**a)** $1\ 000\ 000^{\frac{2}{3}}$

10 000

**b)** $3^{1.9}$, rounded to three decimal places.

8·064

(8·063 ½)

**c)** How many Litres are there in 4.9 kL?

4900 L

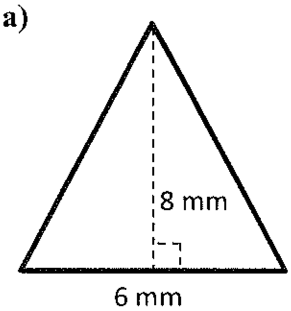
**d)** How many $cm^2$ are there in 0.0059 $m^2$?

59

**7. [8 Marks: 1, 1, 1, 2, 3]**

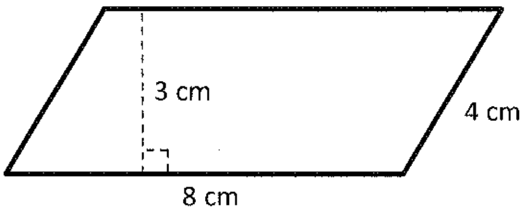
Find the areas of the following shapes, if needed round your answers to 1 decimal place

**a)**



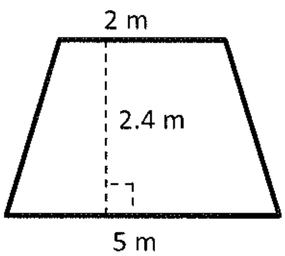

24mm²

**b)**



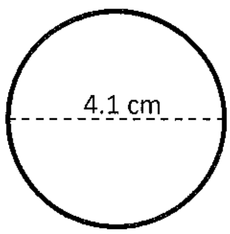

24cm²

**c)**



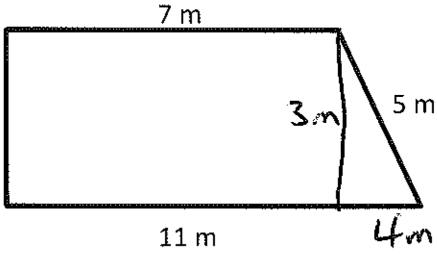

8·4 m²

**d)**




13·2cm²

(✓ if wrote down formula and filled in 4·1 for radius) e)




Hint: Use Pythagoras

height = 3m ✓

Area $= \frac{7+11}{2} \times 3$

$= 9 \times 3$

$= 27 \text{ m}^2$ ✓✓

**8.** **[2 Marks]**
What area of tin is needed to make an open box which is 4 cm deep with a base measuring 8cm by 6 cm?

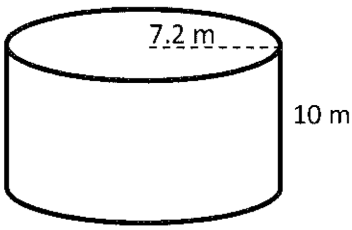
Base = $8 \times 6$ = $48 \text{ cm}^2$

Side 1+3 = $8 \times 4 \times 2$ = $64 \text{ cm}^2$

Side 2+4 = $6 \times 4 \times 2$ = $48 \text{ cm}^2$

$160 \text{ cm}^2$ ✓✓

**9.** **[1 Mark]**
Find the surface area of the cylinder. Round your answer to two decimal places.

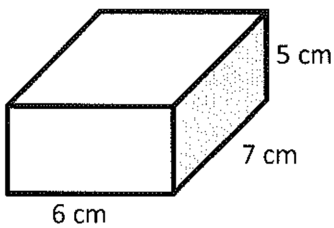



S.A. $= 2\pi r^2 + 2\pi r h$

$= 2 \times \pi \times 7.2^2 + 2 \times \pi \times 7.2 \times 10$

$= 778.11 \text{ m}^2$ ✓

**10.** **[2 Marks]**
Find the __capacity__ of the rectangular container

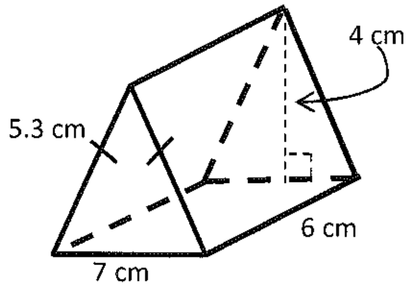



Vol $= 5 \times 7 \times 2$

$= 70 \text{ cm}^3$ ✓

capacity = 70ml ✓

**11. [5 Marks: 2, 3]**
Use the prism below to answer the following questions




**a)** Find the volume of the prism.

$$Vol = \frac{1}{2} \times 4 \times 7 \times 6$$
$$= 84\,cm^3$$ ✓✓

**b)** Find the Surface Area of the prism.

$2\times$ Triangle $= \frac{1}{2} \times 4 \times 7 = 28\,cm^2$

Bottom $= 7 \times 6 = 42\,cm^2$

2 sides $= 5.3 \times 6 = 63.6\,cm^2$

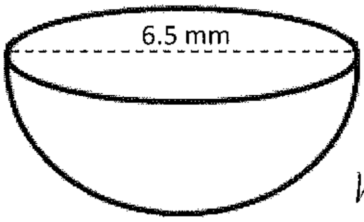
$133.6\,cm^2$ ✓

✓✓ ecf −1 per error

**12. [7 Marks: 3, 4]**

Find the exterior surface area of the following shapes.
Round your answer to 1 decimal place.

**a)**




Hint: this is a hemisphere

Circle $= \pi \times 3.25^2$

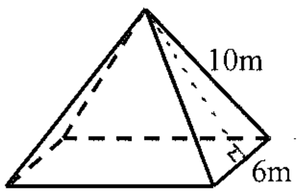
$= 33.1830724$ ✓

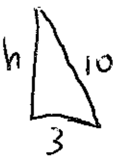
hemisphere $= \frac{1}{2} \times 4\pi \times 3.25^2$

$= 66.3661449$ ✓

Total Surface Area $= 99.5\,mm^2$ ✓

**b)**




$h = \sqrt{10^2 - 3^2}$

$= \sqrt{100 - 9}$

$= \sqrt{91}$ ✓

Base $= 14 \times 6 = 84\,m^2$ ✓

$2 \times$ sm $\Delta = 2 \times \frac{1}{2} \times 6 \times \sqrt{91} = 6\sqrt{91}\,m^2$ (85.85452813)

$2 \times$ big $\Delta = 2 \times \frac{1}{2} \times 14 \times \sqrt{91} = 14\sqrt{91}\,m^2$ (133.5514882)

$274.8\,m^2$ ✓✓

**~ END OF SECTION TWO ~**